

# MNIST Handwritten Digit Recognition

Comparing MLP · CNN · Transformer

Assignment 4.1 | Ziru Xu

# Problem & Dataset

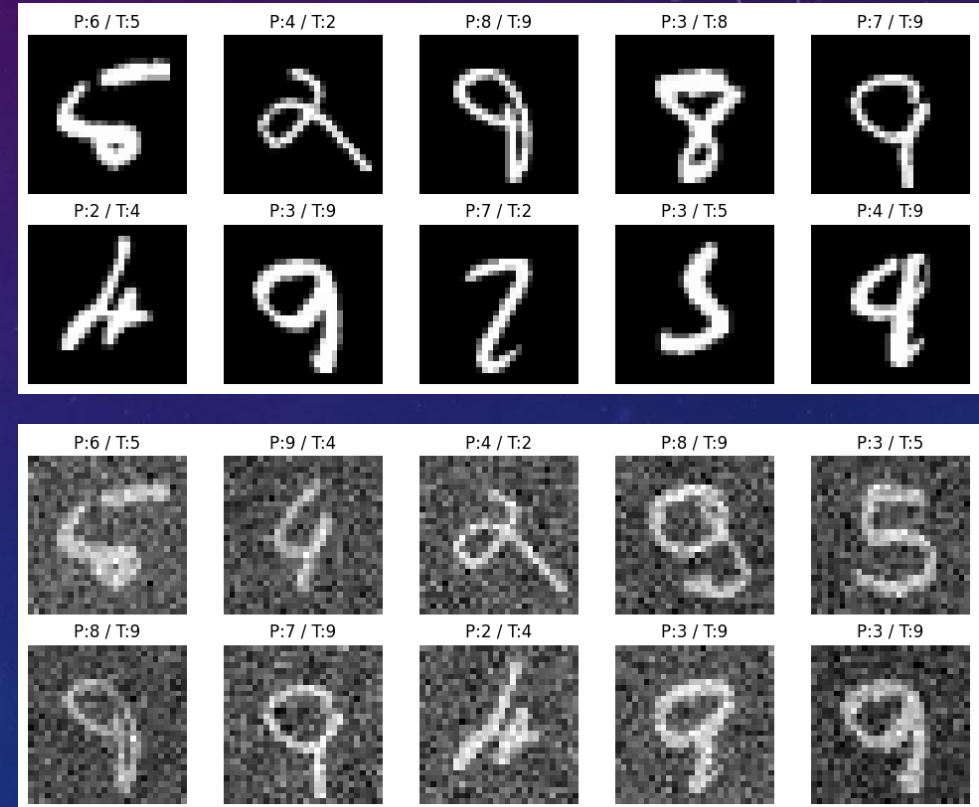
## The Problem

- Classify handwritten digits 0–9
- MNIST: 60,000 train / 10,000 test images
- Image size:  $28 \times 28$  grayscale pixels
- 10 classes (digits 0 through 9)

## Data Preprocessing

- Normalize to  $[-1, 1]$  via mean=0.5, std=0.5
- Noisy split: add Gaussian noise ( $\sigma=0.2$ )
- Batch size: 64 train / 1000 test
- Shuffle training data each epoch

## Sample Digits



Clean vs Noisy

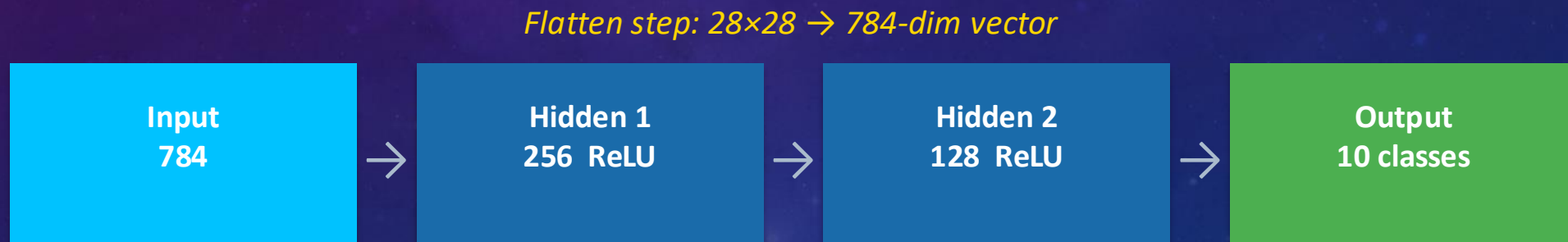
# Model 1 — Multi-Layer Perceptron (MLP)

## Overview

- Shallow fully connected dense network
- Image is flattened:  $28 \times 28 = 784$  input neurons
- No spatial awareness — pixels treated independently
- Simple, fast baseline

## Results

- Clean: 96.94% | Noisy: 96.08%
- Drop under noise: -0.86%



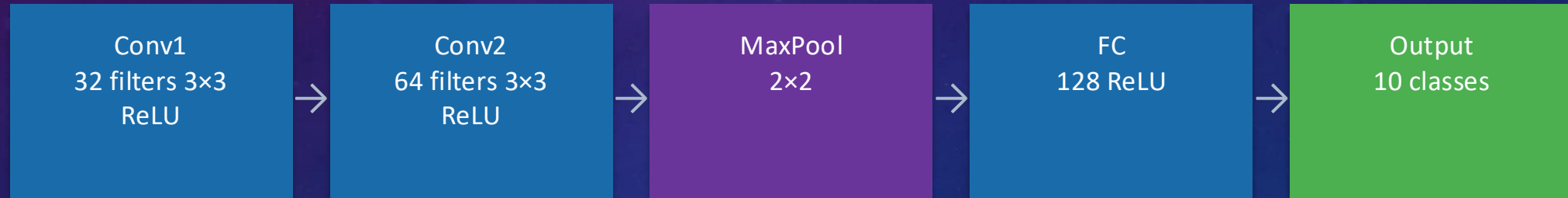
# Model 2 — Convolutional Neural Network (CNN)

## Overview

- Operates directly on 28×28 images
- Learns local features: edges, curves, strokes
- Spatial hierarchy via two conv layers + pooling
- Far fewer parameters than equivalent MLP

## Results

- Clean: 99.01% | Noisy: 98.26%
- Best accuracy — most robust to noise
- Drop under noise: -0.75% (smallest of all)





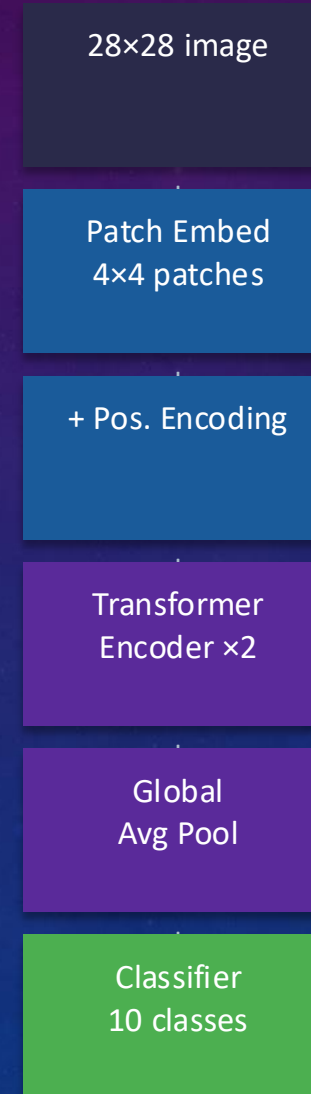
# Model 3 — Transformer Encoder

## Architecture

- Image divided into 4×4 patches → 49 tokens
- Each patch projected to embedding dim = 64
- Learnable positional encoding preserves order
- 2 × TransformerEncoderLayer (4 heads)
- Global average pooling → linear classifier

## Results

- Clean: 97.65% | Noisy: 96.62%
- No conv layers — pure attention mechanism
- Competitive with CNN; slightly less robust



# Model Comparison

| Model       | Clean Acc. | Noisy Acc. | $\Delta$ (drop) | Verdict          |
|-------------|------------|------------|-----------------|------------------|
| MLP         | 96.94%     | 96.08%     | -0.86%          | Solid baseline   |
| Transformer | 97.65%     | 96.62%     | -1.03%          | Strong, no convs |
| CNN ★       | 99.01%     | 98.26%     | -0.75%          | Best overall     |



# Key Takeaways

## CNN is best for image tasks

Spatial inductive bias → highest accuracy + best noise robustness (99.01% clean, -0.75% noise drop).

## Transformers work — even without convolutions

Self-attention captures global context; 97.65% clean accuracy with pure attention on 28×28 images.

## MLP is a strong baseline

96.94% accuracy despite zero spatial awareness — flattening loses structure but MNIST is simple enough.

## Gaussian noise affects all models, but minimally

All models drop < 1.1%; CNN loses the least (-0.75%), showing convolutions learn robust features.